

An AI-Powered Real-Time Environmental Sound Recognition and Alert System

Session Year: 2026-2027

Faculty: MR. TAPAS MISHRA

Team Members

MANAV — 2501920110217

LUCKY — 2501920110210

ABHISEK — 2501920110.

Section: CS IT D2

Contents

- Introduction — environmental sound recognition and why it matters
- Problem Statement — limitations of manual and single-shot sound recognition
- How the Project Works — audio capture, Mel Spectrogram, CNN, confidence and alerts
- How the Project Will Be Made — development phases, training and integration
- Team Division — responsibilities of the three team members
- Literature Survey — research papers, technologies and identified gaps

Introduction

Domain: Artificial Intelligence, Audio Processing and Environmental Sound Classification

What is the topic?

- Environmental sound recognition uses AI to identify non-speech sounds such as knocks, alarms, sirens, dog barks, glass breaking and baby crying.
- Instead of analyzing only spoken words, the system learns acoustic patterns from everyday surroundings.
- For this project, the microphone continuously captures audio in 1-second chunks and the trained model predicts the most likely sound.

Why is it important?

- Accessibility: converts important sounds into visible alerts for users who may not hear them clearly.
- Privacy: local audio processing can avoid sending raw audio to external servers.
- Safety: can detect alarms, sirens, glass break or distress-related sounds.
- Context awareness: lets software react to environmental events instead of only human speech.

Problem Statement

Existing Challenges

- Traditional rule-based audio detection often depends on fixed volume or frequency thresholds and can confuse different sounds.
- Camera-based monitoring creates privacy concerns and is ineffective when visual information is unavailable.
- Cloud inference can introduce internet dependence, transmission delay and additional privacy concerns.
- A single prediction can be uncertain when sounds are short, noisy or similar.

Proposed Solution

- A locally running AI sound classifier based on Mel Spectrograms and a CNN.
- Continuous 1-second audio analysis with a ranked probability list.
- High, medium and low confidence states instead of forcing every sound into a class.
- A 'Provide More Audio' option when more evidence is needed.
- Timestamped event history and configurable visual alerts.

Core Problem

People and software often lack a simple real-time way to recognize important environmental sounds without relying on cameras, manual monitoring, or cloud-based services.

How the Project Works

Software + AI pipeline



AI decision logic

- The microphone captures a 1-second audio chunk.
- The audio is converted into a Mel Spectrogram, a time-frequency image.
- The CNN learns patterns in the spectrogram and outputs class scores.
- Softmax converts scores into probabilities, which are ranked from highest to lowest.
- High confidence: identify sound. Medium: show alternatives + request more audio. Low: show UNKNOWN + request more audio or ignore.

Processing mode: Local / on-device inference for the prototype

How the Project Will Be Made

Development phases

- 1 Requirement & UI Wireframe** Finalize sound classes, confidence states, paper wireframe and Figma prototype.
- 2 Dataset Preparation** Collect/select labeled sounds, clean audio, resample and create 1-second training samples.
- 3 Feature Extraction** Generate Mel Spectrograms from training audio.
- 4 CNN Training** Train, validate and test the custom CNN; measure accuracy, loss and confusion matrix.
- 5 Live Audio System** Capture microphone audio continuously in 1-second windows and send each window to the model.
- 6 Application Integration** Connect AI predictions to dashboard, confidence states, event history and alerts.
- 7 Testing & Demo** Test noise, similar sounds, unknown audio, confidence handling and real-time response.

Project Division Among 3 Team Members

PERSON 1 — TEAM LEADER

AI/ML & System Integration Lead

- Dataset preparation and sound classes
- Mel Spectrogram pipeline
- Custom CNN training and evaluation
- Prediction + confidence logic
- Final integration and architecture

Shared: Git repository, documentation and viva preparation

PERSON 2

Frontend & UX Engineer

- Paper wireframe + Figma prototype
- Dashboard and live waveform UI
- High / Medium / Low confidence screens
- Event History, Sound Classes and Settings UI
- Final presentation UI and prototype

Shared: Git repository, documentation and viva preparation

PERSON 3

Audio / Backend & QA Engineer

- Microphone capture and 1-second buffering
- Event storage / SQLite
- Desktop alerts and notification logic
- Testing under noise and different conditions
- Integration support and QA

Shared: Git repository, documentation and viva preparation

Literature Survey

Format follows the provided college template

S. No	Paper Name	Author Name	Technology Used	Gap / Limitations
1	Environmental Sound Classification with Convolutional Neural Networks (2015)	Karol J. Piczak	Spectrograms + 2-layer CNN + fully connected layers	Focuses on short-clip classification; does not provide a continuous user-facing monitoring and alert workflow.
2	Deep Convolutional Neural Networks and Data Augmentation for Environmental Sound Classification (2017)	Justin Salamon; Juan Pablo Bello	Deep CNN + audio data augmentation	Addresses data scarcity and classification accuracy, but not confidence-driven interaction or additional-audio confirmation.
3	Multi-Temporal Resolution Convolutional Neural Networks for Acoustic Scene Classification (2017)	Alexander Schindler; Thomas Lidy; Andreas Rauber	Parallel CNNs + log-Mel spectrograms at multiple temporal resolutions	More complex architecture aimed at scene classification; a lightweight college application needs simpler real-time deployment.
4	Recurrent Neural Networks for Polyphonic Sound Event Detection in Real Life Recordings (2016)	Giambattista Parascandolo; Heikki Huttunen; Tuomas Virtanen	Multi-label BLSTM RNN	Handles overlapping events, but is more complex than the single-dominant-sound CNN prototype proposed here.
5	CNN Architectures for Large-Scale Audio Classification (2016/17)	Shawn Hershey et al.	Large-scale CNN architectures; AudioSet	Large datasets and models are powerful but heavier than needed for a small local college prototype.

Note: 'Gap / Limitations' identifies the gap between the cited research focus and the features proposed in this project; it does not claim the papers are defective.

Literature Survey References

- 1. Piczak, K. J. (2015). Environmental Sound Classification with Convolutional Neural Networks. IEEE MLSP 2015. DOI: 10.1109/MLSP.2015.7324337.
- 2. Salamon, J., & Bello, J. P. (2017). Deep Convolutional Neural Networks and Data Augmentation for Environmental Sound Classification. IEEE Signal Processing Letters, 24(3), 279-283. DOI: 10.1109/LSP.2017.2657381.
- 3. Schindler, A., Lidy, T., & Rauber, A. (2017). Multi-Temporal Resolution Convolutional Neural Networks for Acoustic Scene Classification. DCASE 2017.
- 4. Parascandolo, G., Huttunen, H., & Virtanen, T. (2016). Recurrent Neural Networks for Polyphonic Sound Event Detection in Real Life Recordings.
- 5. Hershey, S., et al. (2017). CNN Architectures for Large-Scale Audio Classification. ICASSP / AudioSet-related work.
- 6. Piczak, K. J. (2015). ESC: Dataset for Environmental Sound Classification. ACM Multimedia 2015. DOI: 10.1145/2733373.2806390.

Project direction

This project applies the established environmental sound classification pipeline to a simple, local, continuous monitoring application with confidence-aware interaction.